

# Miguel Beltran

(530) 231-2669 | [migueljoaquinbeltran@gmail.com](mailto:migueljoaquinbeltran@gmail.com) | [linkedin.com/in/miguel-j-beltran](https://linkedin.com/in/miguel-j-beltran) | [github.com/migueljbeltran](https://github.com/migueljbeltran) | [migueljbeltran.github.io](https://migueljbeltran.github.io)

## EDUCATION

### University of California, Davis

Davis, CA

*Bachelor of Science in Computer Science*

*Expected June 2027*

**Relevant Coursework:** Algorithms & Data Structures, Operating Systems, Machine Learning, Artificial Intelligence, Computer Architecture

## EXPERIENCE

### ASUCD Picnic Day

Davis, CA

*Assistant Technical Director*

*Nov 2025 – Present*

- Built React/TypeScript features for a production event platform serving 70,000+ UC Davis Picnic Day attendees.
- Implemented onboarding tutorials, calendar exports, and interactive map flows to improve event discovery and attendee navigation.
- Migrated data workflows from Supabase to embedded SQLite with authenticated Google Sheets sync, upserts, and stale-record cleanup.
- Containerized the full stack with Docker Compose, reducing local setup to one command and improving contributor onboarding.

### Include

Davis, CA

*Software Engineer*

*Oct 2025 – Apr 2026*

- Shipped 3 responsive React/Next.js sections for a client website, translating Figma requirements into reusable production components.
- Built prop-driven layouts, optimized Next.js image rendering, and modular SCSS mixins to keep pages maintainable across desktop and mobile.

### Google Developer Student Club

Davis, CA

*Software Engineer*

*Oct 2024 – Jun 2025*

- Built a Python/Flask Slack bot with Slack API webhooks and slash commands to automate club onboarding and announcements.
- Implemented parameter parsing and scheduled announcement workflows, saving organizers 2+ hours per week.

## PROJECTS

### dltracker - Deadlock Analytics Platform | *Next.js 16, TypeScript, PostgreSQL, Redis, Playwright, Vitest, GitHub Actions*

- Built and deployed a full-stack analytics platform used by 1,000+ users, integrating Steam and Deadlock APIs across 10 pages and 45+ reusable components.
- Designed player identity resolution for ambiguous leaderboard records across 5 regions using account metadata and match-history signals.
- Implemented Redis rate limiting, ISR caching, Zod validation, Sentry monitoring, GitHub Actions CI, and Playwright/Vitest tests for production reliability.

### Spotify Hit Classifier - ML Pipeline | *Python, TensorFlow, XGBoost, scikit-learn*

- Built an end-to-end ML classification pipeline over 32,000+ Spotify tracks using scikit-learn, XGBoost, and TensorFlow.
- Engineered artist, genre, and audio features with GridSearchCV tuning, improving Random Forest accuracy to 80%.
- Compared regression and classification approaches, selecting binary hit prediction after feature analysis showed weak regression signal.

### Kitch - Kitchen Management App | *Java, Spring Boot, React, H2, JPA*

- Built a full-stack kitchen management app with a Spring Boot REST API and React/Vite frontend for inventory, shopping lists, and recipes.
- Designed layered backend architecture with controllers, services, repositories, DTOs, and JPA entities across 20+ REST endpoints.

### 8-Bit CPU - Custom Processor | *Logisim, Computer Architecture*

- Designed a 5-stage pipelined 8-bit CPU with hazard detection and data forwarding, executing 20+ custom ISA instructions.

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Python, Java, C, C++, Go, SQL

**Frontend:** React, Next.js, Tailwind CSS, SCSS, HTML/CSS

**Backend:** Node.js, Express, Spring Boot, Flask, FastAPI, REST APIs, Prisma

**Data:** PostgreSQL, SQLite, Redis, Firestore, Supabase

**Testing/DevOps:** Playwright, Vitest, Pytest, Docker, CI/CD, GitHub Actions, Linux, Sentry, Vercel

**ML:** TensorFlow, PyTorch, XGBoost, scikit-learn, Pandas, NumPy